

My teaching philosophy revolves around *constructivism*, which states that “students construct knowledge rather than merely receive and store knowledge transmitted by the teacher” (Ben-Ari, 1998). From constructivists’ perspective, learning is the active process of constructing knowledge through hands-on activities and experiences to expand from existing understanding (Duffy & Cunningham, 1996). It is the students rather than the instructors that take the initiatives of learning, so it is important for instructors to provide sufficient support and encourage students to explore. Therefore, I am committed to 1) building classroom environments where students can feel safe to explore and construct knowledge, and 2) executing instructions that facilitate knowledge construction.

Building Welcoming Classrooms and Adopting Flexible Policies

My goal is to create a classroom climate where students feel comfortable engaging with instructors and peers and can trust that their efforts to take intellectual risks in constructing knowledge will be met with support with no judgement. While this may be an ambitious goal, I believe my experiences as a co-instructor for two CS courses (Computer Architecture and Discrete Structures) at UIUC with disparate course formats and structures have equipped me with reflexive insights and practical approaches to creating a supportive learning environment in various settings.

In Spring 2024, I served as a co-instructor for Computer Architecture, a large (~280 students) required CS course that I had already supported for several semesters on the course staff. This course has been using a flipped classroom model since 2020, with group-based collaborative learning during each class meeting after a mini lecture. When I was on course staff, I had always loved how engaged students were in this structure and how it allowed me to build connections with them even in such a large class, so I continued this structure and implemented it with care when I took over the course. Active, collaborative learning in a large classroom is really challenging, because students may feel that instructors are transferring the responsibility of teaching to the students and leaving them on their own because instructors are now playing the supporting role. Therefore, each day after the mini lecture, I would walk among groups and check with students, asking how they were doing and whether they had questions about the lecture or the collaborative assignments. Initially, many students would say “We’re good. No questions at this point. Wait, actually I didn’t quite understand...” and then bring up their question. Apparently, these students felt shy to ask their questions in front of the whole class or raise their hands during group activities, but when they felt cared and safe with me, they gained the courage to speak up their confusions and ask questions. By the end of the semester, students became comfortable actively seeking help, which was an important factor for success in flipped classrooms. In the anonymous, voluntary course evaluation survey, students commented on my passion and willingness to help, and I believe that my contagious enthusiasm had contributed to the welcoming classroom environment (“Hongxuan was always super helpful and his passion for teaching was very obvious. He would get excited when he would help us and we would understand the topic, which was very refreshing to see. He's awesome!”).

In Summer 2025, I co-taught a smaller (~50 students) course Discrete Structures in a traditional lecture format remotely. Even in this format, I still valued in-class engagement by pausing frequently to check for understanding. I asked students to use Zoom reactions such as thumbs up

and thumbs down during lectures to indicate whether they were following along, and they were also encouraged to ask questions in public and/or private chats. These approaches helped me quickly gauge understanding and adjust the pace of instruction, while also giving quieter students a low-stakes way to participate. Besides, I used breakout room discussions throughout the lectures, when students were placed into smaller groups with course staff support and solved discussion problems. Similar to my practices for Computer Architecture, I actively checked up on students in the breakout rooms instead of passively waiting for them to seek help. By doing so, I got to have more direct interactions with the students, and they became more comfortable speaking up their confusions or asking questions both during the lectures and in breakout rooms.

In addition to building a welcoming classroom that fosters students' engagements and connections with me, I am also interested in course policies that encourage perseverance and reduce attrition. My approach to flexible policies is informed by my own experiences as a student and my teaching experiences, mostly from the Computer Architecture course. To lower the stakes of certain assignments and/or exams and account for unexpected situations, we had policies such as allowing homework to be submitted for partial credit after deadlines, offering second-chance testing opportunities, and providing optional extra practice assignments with extra credit, etc. Many of these policies were made possible and scalable thanks to support of various learning technologies, such as [PrairieLearn](#) (a learning platform with huge flexibility) and our [Computer-Based Testing Facilities](#) at UIUC, and I look forward to exploring what is possible at your school.

In my classroom, I will prioritize student engagement by following those pedagogical practices mentioned above, and I will explore different policies to enhance student experience and lower stress levels. I will also train my course staff to make sure everyone contributes to welcoming interactions with students. In addition, I will actively learn about other approaches and studies in the CS/engineering education research field to further refine my instruction, classroom environments, and course policies.

Understanding and Addressing Student Difficulties in Learning

Whether it is traditional lecturing or active learning, instructors' ability to explain things effectively plays a crucial role in students' understanding. Because instructors and students tend to have different expertise, experiences, and perspectives, it is not always easy for instructors to communicate knowledge from their minds directly to the students. This often leads to the "expert blind spot", where instructors fail to recognize student difficulties, resulting in ineffective teaching (Nathan et al., 2001). Aspiring to be an effective instructor, I am interested in investigating student difficulties and addressing them with my research skillset.

In my everyday teaching, I pay attention to where students struggle and react accordingly. When co-instructing Discrete Structures, I proposed and executed the idea of reviewing students' discussion problem submissions and compiling a list of common mistakes with comments and tips to help students prepare for their weekly exams. Students appreciated knowing that their difficulties were shared by others and valued the tips we provided for addressing them. I will continue documenting and analyzing student difficulties, which I hope to address by providing more explanations and alternate perspectives and/or conducting more rigorous research.

My research in computing education also centers on uncovering student difficulties and addressing them through evidence-based methods. For example, in our work on graph layering, a data modeling technique for algorithm design, we 1) conducted think-aloud interviews to identify students' conceptual challenges, 2) designed scaffolded problems and reading materials to provide targeted support, and 3) evaluated effectiveness of these approaches through experiments. I will continue using this “identifying challenges — designing solutions — examining effectiveness” pipeline to address more student difficulties in the future.

Looking Ahead

I will continue to ground my teaching in constructivist principles by creating environments where students can safely take intellectual risks and actively construct knowledge. I also aim to refine my use of evidence-based practices that enhance engagement, reduce attrition, and promote equity in computer science classrooms, while mentoring course staff and collaborating with colleagues on instructional innovations. Besides, I will continue to apply my research skills to better identify and address student difficulties in learning. I plan to use diagnostic assessments, student feedback, and evidence-based scaffolding strategies to uncover common misconceptions and provide targeted support. Through these efforts, I strive to ensure that every student leaves my courses with both a solid understanding of the material and the confidence that their learning efforts will be recognized and supported.

References

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